

12 (long)

(a) Discuss the *leak-before-break* criterion in connection with pressure vessel design. [6]

(b) A steel thin-walled spherical pressure vessel is to be used in a chemical processing plant. The vessel has a wall thickness of 10 mm and a diameter 2 m and is required to operate safely at a pressure of 3 MPa. The steel was specified with a fracture toughness K_{IC} of $85 \text{ MPa}\sqrt{\text{m}}$.

Non-destructive testing revealed that in the finished vessel there are small cracks in the inner surface with their planes normal to the wall surface. The stress intensity factor K for such cracks is given by

$$K = 1.13\sigma\sqrt{\pi a}$$

where σ is the tensile stress in the wall and a is the crack depth.

(i) Show that a leaking vessel satisfies the leak-before-break criterion. [6]

(ii) Under fatigue loading by a stress intensity range ΔK , the cracks grow at a rate

$$\frac{da}{dN} = A\Delta K^4$$

where $A = 5 \times 10^{-16} (\text{m/cycle})(\text{MPa}\sqrt{\text{m}})^{-4}$ and N is the number of fatigue cycles. Estimate how many cycles it will take for leakage to occur if the steel contains a crack of depth 2 mm. [8]

(c) Consider a thin-walled spherical pressure vessel. For a fixed vessel radius, derive a *material merit index* in terms of the fracture toughness K_{IC} and the yield stress of the wall material σ_y , which should be maximised in order to operate at a maximum pressure consistent with the leak-before-break criterion. It is required that plastic deformation should not occur. Use the data in the table below to select the best material according to the merit index you have derived.

Material	σ_y (MPa)	K_{IC} ($\text{MPa}\sqrt{\text{m}}$)
Medium carbon steel	603	75
Titanium alloy	748	65
Aluminium alloy	265	30
Magnesium alloy	235	20

[10]

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Answers

1. (b) $v(x) = x \tan \theta \left(1 - \frac{x}{2L}\right)$
2. (a) $t = \frac{LT}{EA}$, $s = \frac{5L^3}{3EI}$
(b) $h = \frac{HL^3}{E(AL^2 + 3I)}$, $v = \frac{VL^3}{E(AL^2 + 3I)}$
3. (b) External force at E is 1 to the right, horizontally.
4. (a) $I_{tube} = 16.9 \times 10^{-6} \text{ m}^4$, $I_{bar} = 8.53 \times 10^{-6} \text{ m}^4$
(b) 10.5 mm
5. (a) (i) 0 , 0 , 600 , -700 kN
(ii) 0 , -600 , 0 , 608 kN
(iii) -700 , -700 , 849 kN
(b) 1.98 m
6. (a) $R_R = 3 \times 10^6 \text{ Nm}^{-1}$, $R_L = 15 \times 10^6 \text{ Nm}^{-1}$
(b) $S = (2x - 0.015x^2) \text{ MN}$, $M = (x^2 - 0.005x^3) \text{ MNm}$
(c) $t > 21.2 \text{ mm}$
8. (b) Thickness of metal lost on either side 0.33 mm (< 1 mm). Material can be used.
9. (a) $D_B/D_A = 0.414$
(b) Theoretical density = 3.21 Mg m^{-3}
11. (b) (i) $\epsilon_t = 0.5$ (ii) $\sigma_n \cong 107 \text{ MPa}$
(c) $N_f \cong 6.25 \times 10^4$ cycles
12. (b) (ii) $N_f \cong 9.81 \times 10^7$ cycles
(c) K_{IC}^2/σ_y , Best material: medium carbon steel